Anshaditya Sharma

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Education

Vellore Institute of Technology (VIT) Chennai — Chennai, Tamil Nadu
B.Tech in Computer Science and Engineering; CGPA: 8.15 / 10.0 — *Aug 2023 – May 2027*

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, System Design, Machine Learning, Applied Cryptography

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, C++, C, Go, SQL, HTML/CSS
Backend & Distributed Systems: Node.js, Express.js, FastAPI, REST APIs, WebSockets, Concurrency, Multithreading, System Design
Databases & Infrastructure: PostgreSQL, MongoDB, Redis, Firebase/Firestore, Docker, GitHub Actions, Linux/Unix Shell Scripting
ML & Security: PyTorch, Scikit-learn, TensorFlow, NLTK, NumPy, SHAP, Streamlit, Isolation Forest, Anomaly Detection, JWT Auth, Applied Cryptography
Developer Tools: Git, GitHub, Postman, Swagger/OpenAPI, Unit Testing, Microservices, Agile/Scrum

Key Projects

bb – Live Paper-Trading Exchange & Matching Engine — Code
Node.js, TypeScript, PostgreSQL, WebSockets, System Design — Feb 2026

- Engineered a zero-dependency order-matching engine processing 2.1M operations/sec at 1.2μs p99 submit latency using price-time priority, sorted price levels, intrusive FIFO queues, and O(1) cancellation maps.
- Architected a modular monolith streaming sequenced order-book deltas and trade tapes over WebSockets/REST, with write-behind PostgreSQL persistence and reservation-based cash/asset models enforcing zero-shorting.
- Integrated an execution-log-grounded LLM fill-explainability engine and automated market-maker/noise-trader liquidity bots; deployed on Railway with serverless PostgreSQL.

Behavioral Anomaly Detection System — Code
Python, Scikit-learn, PyTorch, SHAP, Streamlit — Jan 2026

- Built an Isolation Forest, Random Forest, and GRU ensemble over 22 behavioral features and 30-day rolling baselines to detect brute force, credential stuffing, lateral movement, impossible travel, and low-and-slow exfiltration.
- Evaluated 113K+ synthetic access sessions across 370 entities using time-based train/test splitting, achieving 0.899 PR-AUC and 99.7% precision at a 1% alert budget.
- Added SHAP explanations, cold-start scoring, and a Streamlit analyst dashboard for ranked alerts, entity history, and feedback.

LectureFind – AI-Powered YouTube Lecture Navigator — Code
React, FastAPI, Python, NLTK, Chrome Extension — Nov 2025

- Built a full-stack platform that searches concepts within YouTube transcripts and returns clustered timestamp ranges for direct in-video navigation.
- Developed a FastAPI NLP backend using transcript retrieval, WordNet synonym expansion, Wu-Palmer similarity, 45-second temporal clustering, and top-5 related-topic fallback; shipped a React/Vite app and Manifest V3 extension sharing one backend.

Semi-Autonomous VLM-Controlled Rover — Code
ESP32-CAM, Arduino C++, Gemini Vision API, Embedded Systems — Sep 2025

- Designed a hybrid reactive/deliberative control architecture on ESP32-CAM with a 20–50ms on-device safety loop for obstacle avoidance independent of network latency.
- Combined local Whisper speech-to-text and regex intent matching for voice commands, invoking Gemini Vision asynchronously for open-ended queries and structured JSON navigation control.

Patents

- **Co-Inventor**, Indian Patent Application No. 202641075636 – *"Multi-Module Flying Insect Control Device"* (Published Jun 2026), developed with VIT Chennai research faculty and hardware engineering team.

Achievements & Leadership

- **Competitive Programming**: Solved 175+ Data Structures & Algorithms problems on LeetCode (Java, C++); achieved 2-star rating on CodeChef.
- **Hackathons & Innovation**: Secured top positions in national and university hackathons across real-time software and hardware-software integration tracks, leading teams of 4–6 members.
- **Leadership & Mentorship**: Served as Cultural Lead for UP Cultural Club, organizing events for 250+ attendees; conducted group-learning sessions on DSA, System Design, Git, and REST APIs for junior students.